Prototype-CNN for Few-Shot Object Detection in Remote Sensing Images

Gong Cheng, *Member, IEEE*, Bowei Yan, Peizhen Shi, Ke Li, Xiwen Yao, *Member, IEEE*, Lei Guo, and Junwei Han

***Abstract*— Recently, due to the excellent representation ability of convolutional neural networks (CNNs), object detection in remote sensing images has undergone remarkable development. However, when trained with a small number of samples, the performance of the object detectors drops sharply. In this article, we focus on the following three main challenges of few-shot object detection in remote sensing images: 1) since the sample number of novel classes is far less than base classes, object detectors would fail to quickly adapt to the features of novel classes, which would result in overfitting; 2) the scarcity of samples in novel classes leads to a sparse orientation space, while the objects in remote sensing images usually have arbitrary orientations; and 3) the distribution of object instances in remote sensing images is scattered and, therefore, it is hard to identify foreground objects from the complex background. To tackle these problems, we propose a simple yet effective method named prototype-CNN (P-CNN), which mainly consists of three parts: a prototype learning network (PLN) converting support images to class-aware prototypes, a prototype-guided region proposal network (P-G RPN) for better generation of region proposals, and a detector head extending the head of Faster region-based CNN (R-CNN) to further boost the performance. Comprehensive evaluations on the large-scale DIOR dataset demonstrate the effectiveness of our P-CNN. The source code is available at https://github.com/Ybowei/P-CNN.**

***Index Terms*— Convolutional neural network (CNN), few-shot object detection, prototype-CNN (P-CNN), remote sensing images.**

I. Introduction

As REMOTE sensing technology advances, recent years have witnessed a tendency that the quality and quantity of remote sensing images are increasing significantly [1]–[6]. Object detection is a challenging problem in the field of remote sensing image analysis and processing, which serves as a fundamental part for various applications [7].

With the rise of deep learning, especially deep convolutional neural networks (CNNs), a series of outstanding object detection algorithms has emerged [8]–[14] in natural scene images resorting to their powerful feature extraction ability. Note that we adapt the terminology "natural scene images" from these papers [4], [5]. Obviously, unlike natural scenes, remote sensing images usually exhibit the characteristics of arbitrary orientations, highly scattered object occurrence, and complex backgrounds, as shown in Fig. 1. To deal with these problems, a lot of approaches have been put forward and have achieved desirable results [15]–[24]. Despite the excellent performance that CNNs have achieved so far, large-scale datasets are necessarily required.

Recently, few-shot object detection has received increasing attention in natural scene images [25]–[31], which means learning concepts from limited training samples with annotation. Specifically, given an auxiliary set of base classes with abundant annotated examples and novel classes with only few annotated samples, the few-shot detectors can detect the objects of both base and novel classes. It is worth noting that these methods were mainly designed for natural image datasets such as the PASCAL VOC dataset, where the orientation of the objects is relatively unitary and most of the object instances locate in the middle of the images (see Fig. 1). As a comparison, the objects in remote sensing images have arbitrary orientations and locate at quite scattered positions. In addition, it is a nontrivial problem to identify foreground objects from the complex background environment in remote sensing images. Therefore, those methods for natural scene images are not suitable for direct application to remote sensing images in a few-shot scenario.

To tackle the aforementioned challenges, in this article, we propose a novel prototype-CNN (P-CNN). P-CNN extends the widely used general object detection framework of Faster region-based CNN (R-CNN) [9]. It is mainly composed of a prototype learning network (PLN), a prototype-guided region proposal network (P-G RPN), and a detection head. In brief, the PLN module learns good prototype representations from support images. The P-G RPN guides better proposal generation by fusing the information of prototypes, and thus, it can identify foreground objects from the complex background. Here, it is noted that the general object detection

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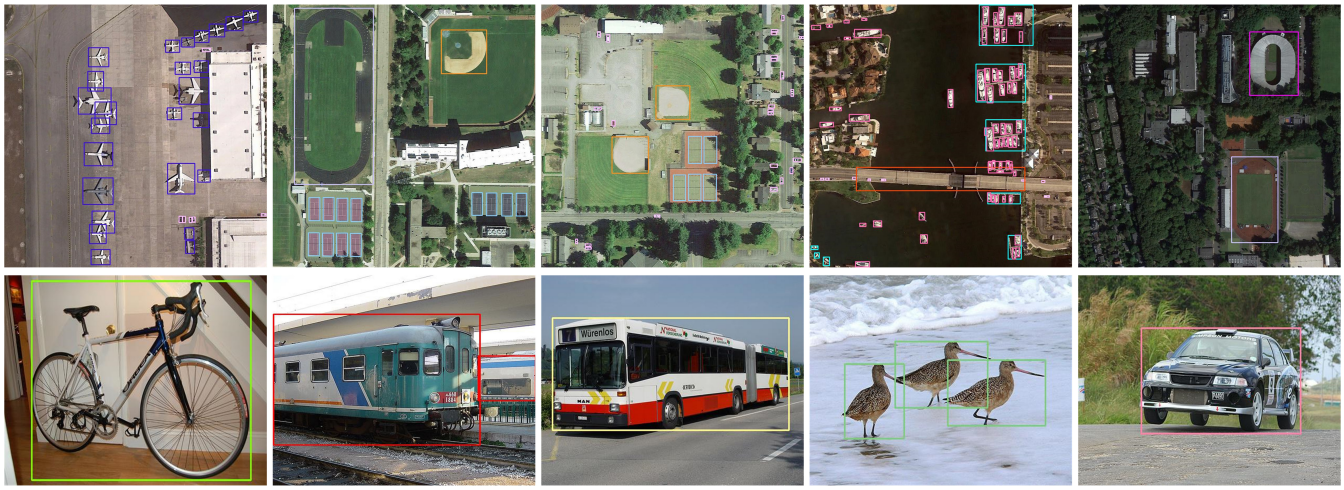


Fig. 1. Two different kinds of images. (Top) Remote sensing images from the DIOR dataset. The objects have arbitrary orientations and locate at quite scattered positions. (Bottom) Natural scene images from the VOC dataset. The orientation of the objects is relatively unitary, and most of the objects locate in the middle of the images. Here, the ground truths are indicated using rectangles and each color represents an object class.

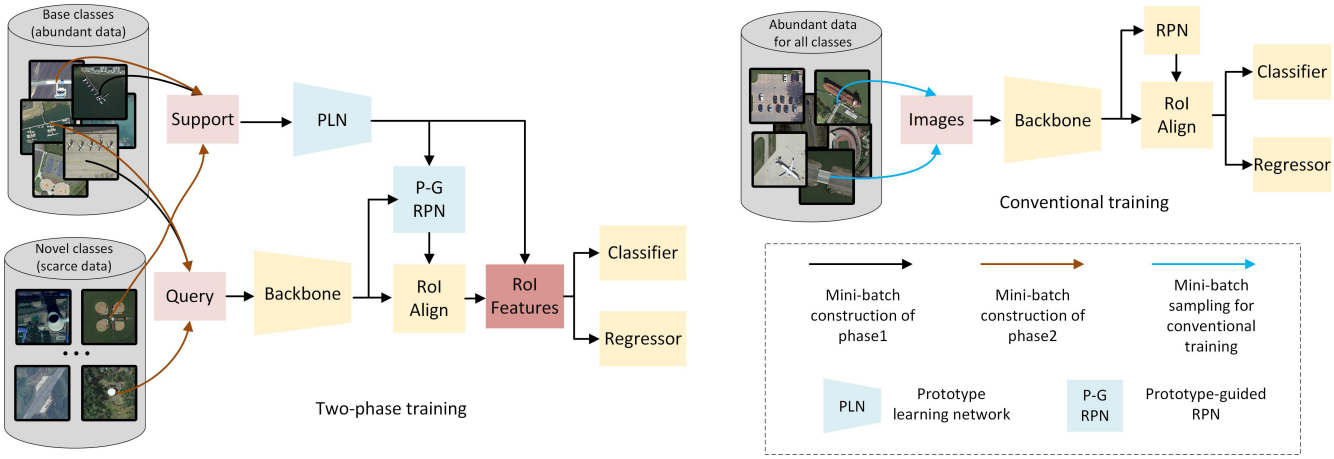


Fig. 2. Illustrations of two different training strategies. (Left) Our two-phase training strategy. In the first phase, the mini-batch composed of support set and query set is obtained from the base dataset containing abundant annotated images. In the second phase, both base and novel categories are utilized to construct the mini-batch. (Top Right) General training strategy. The mini-batch is sampled from the dataset containing abundant annotated images for all categories.

training strategy based on mini-batch stochastic gradient descent (SGD) may not fit for few-shot task. Thus, we use a two-phase training strategy to enable our model to quickly learn the concepts of novel classes, as shown in Fig. 2, and we will detail it in Section IV.

To sum up, our contributions are fourfold. First, we propose a novel P-CNN framework designed for few-shot object detection in remote sensing images. Second, our proposed method leverages PLN to provide class-aware prototypes for guidance and utilizes prototype-guided RPN to highlight the regions for foreground objects, which can provide better proposals for the detection head. Third, we take the issue of orientation variations into consideration through a rather simple multiorientation sample augmentation operation, thus enhancing the model's capability to handle the arbitrary orientation problem of objects. Fourth, we build four different novel/base split settings for few-shot detection task for the first time, on our large-scale DIOR dataset, and exhibit baseline results for further research.

The following is the organization of the remaining parts of this article. Section II briefly reviews the related work. Section III provides preliminary knowledge about our method. Section IV describes our proposed P-CNN method in detail. Comprehensive experimental results are shown in Section V. We conclude this article in Section VI.

II. Related Work

A. Object Detection in Natural Scene Images

Object detection is a fundamental yet challenging task in the field of computer vision. Lately, deep learning-based methods have been dominating this area, which can be divided into two branches: proposal-based methods [8], [9], [32] and proposal-free methods [10], [11], [33], [34]. R-CNN series are the masterpieces of proposal-based methods, which divide detection tasks into two stages. In the first stage, R-CNN [32] and Fast R-CNN [8] utilize selective search [35] to generate region proposals. Then, in the second stage, R-CNN

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applies a support vector machine (SVM) classifier to classify these proposals, and Fast R-CNN proposes to use the softmax classifier to boost the performance. To further accelerate the process, Faster R-CNN designs a lightweight network called RPN to generate region proposals, which improves both efficiency and accuracy. In contrast, one-stage proposal-free methods extract deep features of images using deep CNN and then directly predict the classification scores and bounding-box regression without providing proposals. You Only Look Once (YOLO) [33] is the pioneer work of proposal-free methods. Redmon and Farhadi [10], Liu *et al.* [11], and Redmon and Farhadi [34] made considerable improvements in data augmentation, multiscale training, and some other aspects. In general, proposal-based methods are superior in accuracy, whereas proposal-free methods are faster in speed.

B. Object Detection in Remote Sensing Images

Along with the rapid development of the algorithms mentioned above, many methods are proposed for object detection in remote sensing images. To achieve fast detection for tiny objects in large-scale remote sensing images, Pang *et al.* [36] proposed a unified and self-reinforced framework. To achieve rotation-invariant detection, many research works focus on detecting arbitrary-oriented objects with horizontal bounding-box annotations. For example, Cheng *et al.* [15] proposed a rotation-invariant CNN (RICNN), which contains a new rotation-invariant layer. Li *et al.* [17] proposed a novel RPN including additional multiangle anchors to deal with the multioriented objects. Moreover, Cheng *et al.* [16], [37] proposed a rotation-invariant and Fisher discriminative CNN to simultaneously address the challenges of arbitrary orientation, within-class diversity, and between-class similarity. Since the horizontal bounding boxes are unable to finely capture the real orientations of the objects, there are plenty of works [18]–[20] aiming at detecting objects with oriented bounding-box annotations. For instance, Yang *et al.* [19] introduced an improved smooth L1 loss to handle the boundary problem for the rotating bounding box. Xu *et al.* [20] proposed a novel regression approach by gliding the vertex. Besides, there also exist some methods toward multiscale object detection in remote sensing images [23], [24], [38], [39].

C. Few-Shot Object Detection

Due to the broad application prospects, there are some tentative works on few-shot object detection in natural scene images [25]–[31], [40]. These few-shot object detection methods can be mainly classified into three streams: 1) metric-learning-based methods [26], [30]; 2) fine-tuning-based methods [25], [31]; and 3) meta-learning-based methods [27]–[29].

For metric-learning-based methods, Karlinsky *et al.* [26] proposed the RepMet to jointly train backbone, embedding space, and the representatives for each class, in which the distances between the embedding space vectors and the class representatives are computed to classify the objects. To explore the correlated clues of query and support images, Hsieh *et al.* [30] proposed to impose coattention and coexcitation operations on the deep support and query features. For fine-tuning-based methods, their insights are quite simple, but they can still achieve good performance with the proper implementation. For example, Wang *et al.* [31] proposed an extremely simple fine-tuning approach, which only fine-tunes the last detection layer of the model. As for meta-learning-based methods, Wang *et al.* [27] proposed a MetaDet that disentangles the category-agnostic and category-specific components and uses a weight prediction metamodel to predict the parameters for category-specific layers from few examples. Kang *et al.* [28] extended the widely used proposal-free detection framework YOLOv2 with a reweighting module that reweights the query features with the help of corresponding support images. Similarly, Yan *et al.* [29] followed the same idea to improve the proposal-based method Faster R-CNN, which introduces an extra meta loss for the object attentive vectors and remodels the R-CNN predictor head.

Although few-shot object detection in natural scene images has recently gained more and more attention, there exist rather few methods [41] aiming at few-shot object detection in remote sensing images. Deng *et al.* [41] extended the method proposed in [28] by using feature pyramid network (FPN) to enable multiscale object detection. However, in this work, we achieve the task of few-shot object detection in remote sensing images from different aspects. The main motivations of our method are to obtain better object prototypes and generate high-quality region proposals from the complex background and thus to improve the detection accuracy.

III. Preliminary Knowledge

Our method mainly focuses on extending the Faster R-CNN to adapt the task of few-shot object detection by introducing better prototypes to guide high-quality generation of proposals. Hence, in this section, some preliminary knowledge is introduced in brief, which includes the problem formulation of our few-shot object detection task and prototypical networks [42].

A. Problem Formulation

We follow the common settings for few-shot object detection [28], [29]. Suppose that we are provided with dataset $D_{\text{base}} = \{(x_{\text{base}}, y_{\text{base}}), x_{\text{base}} \in I_{\text{base}}, y_{\text{base}} \in L_{\text{base}}\}$ for base classes C_{base} and dataset $D_{\text{novel}} = \{(x_{\text{novel}}, y_{\text{novel}}), x_{\text{novel}} \in I_{\text{novel}}, y_{\text{novel}} \in L_{\text{novel}}\}$ for novel classes C_{novel}, where $C_{\text{base}} \cap C_{\text{novel}} = \phi$, x are the images in the dataset, y are the corresponding annotations, I stands for the image space, and L stands for the annotation space. We first need to use D_{base} to train a base detector, which can quickly transform the metaknowledge from base classes to novel classes. As shown in Fig. 2, different from the mini-batch construction of general detection algorithms, the inputs for P-CNN consist of support instance patches and query images to be detected. Formally, in each iteration also known as episode, the inputs are composed of (S_i, Q_i), where S_i stands for the instance patch set of support images and Q_i means the query set (for more detailed information of support and

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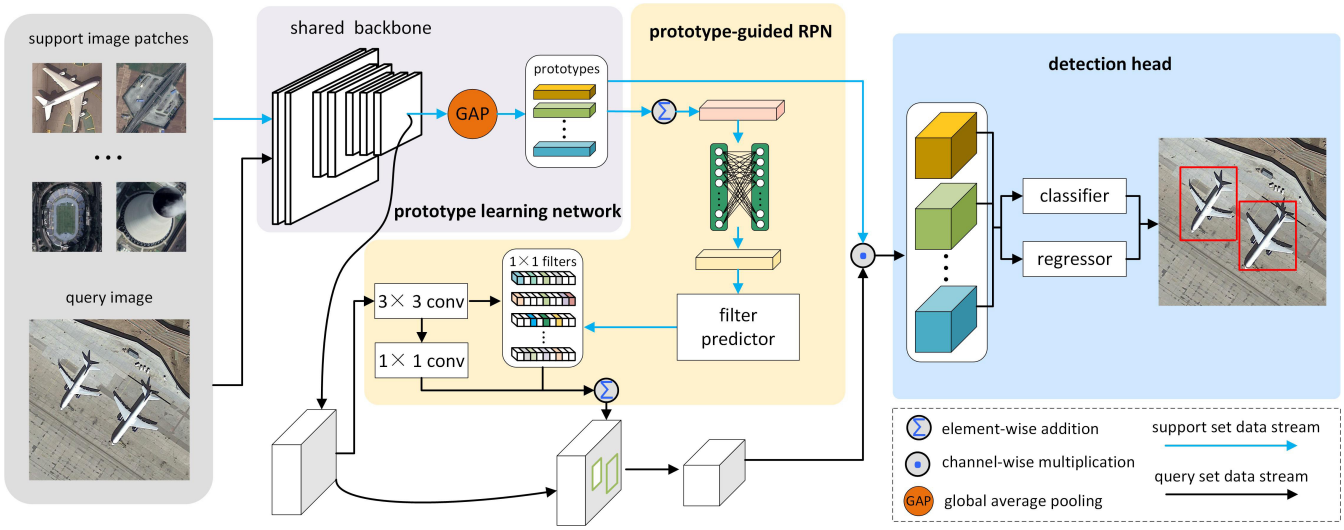


Fig. 3. Framework of our proposed P-CNN method, which mainly consists of three components: a PLN that converts support images into class-aware prototypes, a prototype-guided RPN that injects the information of prototypes into the candidate box scores to guide better generation of object proposals, and a detection head that extends the head of Faster R-CNN to further boost the performance.

query set acquisition, see the training strategy in Section IV). In testing, since we simultaneously detect all categories, D_{test} contains both base and novel categories $C_{\text{test}} = C_{\text{base}} \cup C_{\text{novel}}$.

B. Prototypical Networks

Prototypical networks [42] are originally designed for few-shot classification on the basis that each class can be represented by the mean of its examples in an embedding space. Specifically, (1) formulates the operations to obtain the prototype of class C_k from a collection of annotated data examples D_{C_k} of this category

$$\boldsymbol{p}_k = \frac{1}{|D_{C_k}|} \sum_{(x_i, y_i) \in D_{C_k}} f_\phi(x_i) \tag{1}$$

where $\boldsymbol{p}_k$ is the prototype of class C_k and $f_\phi(\cdot)$ is the feature extractor to encode the input image x_i to an embedding space.

To obtain the classification result of image x, prototypical networks compute the distance between the feature vector of x and $\boldsymbol{p}_k$. Then, the probability of x belonging to class C_k is the softmax result over these distances as follows:

$$p(y = C_k|x) = \frac{\exp(-d(f_\phi(x), \boldsymbol{p}_k))}{\sum_i \exp(-d(f_\phi(x), \boldsymbol{p}_i))} \tag{2}$$

where $d(\cdot)$ is the distance function measuring the similarity between the query features and the prototypes.

In our method, instead of directly calculating the distance between the prototypes and the query features, we use the prototypes as guidance to guide both the generation of proposals and the detection results.

IV. Proposed Method

A. Overview of the Proposed Method

As shown in Fig. 3, our proposed P-CNN is mainly composed of three parts: a PLN, a prototype-guided RPN (P-G RPN), and a detection head. This approach is based on the widely used object detection framework of Faster R-CNN. The images in both query set and support set share the same backbone as the feature extractor. The backbone feature extractor is to embed images into feature maps. Then, motivated by [29] and [30], global average pooling (GAP) [42], [43] is applied to the support features to obtain the prototypes for each class by taking the average of each feature map along the spatial dimension. To integrate the information of all foreground categories, the channelwise summation over these prototypes is utilized, and after that, a fully connected layer follows to remodel these features. We predict the weights for RPN by introducing a lightweight filter predictor that consists of three fully connected layers. With regard to query images, we adopt P-G RPN that takes prototypes into consideration to get more reliable proposals. Furthermore, to encode the information of prototypes and refine features, channelwise soft attention is applied on the output feature maps of region of interest (RoI) Align, and the final detection results are based on these features.

B. Prototype Learning Network

Our proposed P-CNN relies on PLN to obtain the prototypes for each class. Since it is often the case that the objects in the remote sensing images only account for a small proportion of the entire images, thus, instead of concatenating each three-channel RGB image with an additional one-channel binary mask for the object bounding boxes as inputs such as [28] and [29], we directly crop object instance patches according to the ground-truth bounding boxes. The object patches of support images are further resized to 224×224 for input.

In addition, to handle the problem of orientation variance of objects, we designed a very simple data augmentation approach called multiorientation rotation to enrich the orientation space. It rotates each object instance of support images for five times with random angles. Together with the original one, there are six different orientations for each object instance.

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TABLE I
FOUR DIFFERENT BASE/NOVEL CLASSES SPLIT SETTINGS FOR OUR EXPERIMENTS

# split	Novel classes					Base classes
1	baseball field (C1)	basketball court (C2)	bridge (C3)	chimney (C4)	ship (C5)	rest
2	airplane (C6)	airport (C7)	expressway toll station (C8)	harbor (C9)	ground track field (C10)	rest
3	dam (C11)	golf course (C12)	storage tank (C13)	tennis court (C14)	vehicle (C15)	rest
4	expressway service area (C16)	overpass (C17)	stadium (C18)	train station (C19)	windmill (C20)	rest

Specifically, we use PLN to learn embedding space where distinct prototypes for different object classes are encoded. To ease the computation burden and reduce the model parameters to avoid overfitting, we leverage a shared backbone network for both support image patches and query images as the feature extractor $f_\phi(\cdot)$

$$\boldsymbol{p}_i = \mathrm{GAP}\left(f_\phi\left(\mathrm{x}_i^{\mathrm{sup}}\right)\right) \tag{3}$$

$$\mathbf{f}_{\mathrm{que}} = f_\phi\left(x_j^{\mathrm{que}}\right) \tag{4}$$

where x_i^{sup}, GAP$(\cdot)$, and $\mathbf{f}_{\mathrm{que}}$ represent the ith support image patch, GAP, and the extracted deep features of the jth query image, respectively.

Besides, to obtain more representative prototypes, we impose an additional constraint on these prototypes by introducing a simple cross-entropy loss. To this end, the prototypes are encouraged to fall into their corresponding categories.

C. Prototype-Guided RPN

Intuitively, the final detection results depend largely on the quality of the proposals produced by RPN. Since the model is trained on the large-scale base dataset D_{base}, RPN is prone to give the proposals associated with the categories in C_{base} higher scores, which would deteriorate the overall performance. To obtain better foreground candidates, we propose the P-G RPN module that fuses the prototypes and remodels the representations of query images. Specifically, as shown in (5), we do a simple elementwise addition among all prototypes. Then, a fully connected layer is applied to reconstruct the summation of prototypes $\boldsymbol{p}_{\mathrm{fore}}$. Moreover, the filter predictor is utilized to predict the weights $\mathbf{w}$ for the classification layer of RPN, which is formulated in (6)

$$\boldsymbol{p}_{\mathrm{fore}} = \sum_k \boldsymbol{p}_k \tag{5}$$

$$\mathbf{w} = F(fc(\boldsymbol{p}_{\mathrm{fore}})) \tag{6}$$

where $F(\cdot)$ is the filter predictor taht consists of three fully connected layers. As shown in the light-yellow area in Fig. 3, apart from the 3×3 conv and 1×1 conv from the original RPN, we take the weights predicted by the filter predictor as the parameters of the complementary classifier. Then, we fuse the original RPN classification branch and the complementary branch. The two scores are added to form the final classification output. The loss for our P-G RPN remains the same as RPN [9].

D. Detection Head

Despite proposed in Mask R-CNN [44], RoI Align has been widely used in Faster R-CNN with its better feature alignment ability compared to RoI Pooling. After obtaining the features of RoIs using RoI Align, we modify the second stage of Faster R-CNN (or the detection head) to further classify the specific categories of proposals and refine bounding-box regressions.

The original Faster R-CNN head (H) takes the RoIs' feature maps $\mathbf{f}_{\mathrm{rois}} \in {}^{n_1 \times n_2}$ as input and outputs the probabilities over C object classes plus one "background" class and the regressions for each foreground object (n_1 denotes the number of channels for RoI feature maps and n_2 denotes the number of RoIs). Specifically, H consists of a pooling layer and a sequence of fully connected layers that finally end up with two sibling output layers (one for classification and the other for regression). To obtain better feature representations, we replace $\mathbf{f}_{\mathrm{rois}}$ with C class-specific features $\mathbf{f}_{\mathrm{rois}}^k (k = 1, 2, \ldots, C)$ using the prototypes $\boldsymbol{p}_k$ for C classes

$$\mathbf{f}_{\mathrm{rois}}^k = \boldsymbol{p}_k \otimes \mathbf{f}_{\mathrm{rois}} \tag{7}$$

where $\otimes$ is the channelwise multiplication, which acts as a channelwise soft attention for learning class-aware features. After acquiring the remodeled features for each RoI, these features are fed into the detection head of Faster R-CNN. The final classification and regression are conducted based on $\mathbf{f}_{\mathrm{rois}}^k$.

E. Training Strategy

Similar to common few-shot frameworks [29], [45], the pipeline for training our proposed method can be divided into two phases. During the first phase, we only train our model on D_{base} to obtain a basic detector. As shown in Fig. 2, we randomly sample a set of images and then crop object instances from D_{base} to construct our support set. During the second phase, to enable the model to simultaneously detect novel and base classes, we construct a dataset, which contains the whole novel dataset D_{novel}, and a subset D_{subbase}, which is randomly sampled from D_{base}.

V. EXPERIMENTS

A. DIOR Dataset and Evaluation Metrics

In the experiments, we evaluate our proposed P-CNN method on a large-scale publicly available dataset named DIOR [1], which is designed for remote sensing image object detection. This dataset contains 20 common object categories and the details can be found in Table I. It includes totally 23 463 optical remote sensing images that are divided into three parts, namely training set, validation set, and testing set.

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TABLE II

FEW-SHOT DETECTION RESULTS FOR THE NOVEL AND BASE CLASSES ON THE DIOR DATASET FOR THE FOUR BASE/NOVEL SPLITS. WE USE MAP AS EVALUATION MEASURE AND THE BEST RESULTS ARE INDICATED IN BOLD FONT

		Split 1		Split 2		Split 3		Split 4	
#shots	Method	Novel Classes	Base Classes	Novel Classes	Base Classes	Novel Classes	Base Classes	Novel Classes	Base Classes
3	Faster R-CNN [9]	0	45.9	0	45.7	0	51.0	0	48.4
	RepMet [26]	11.3	47.8	5.2	**49.6**	5.5	48.7	10.9	47.4
	FeatReweighting [28]	7.5	**49.4**	4.8	48.5	7.8	45.5	3.7	48.2
	TFA [31]	11.6	47.7	0	48.8	11.1	**51.9**	9.4	49.0
	Meta R-CNN [29]	14.0	46.3	10.5	48.2	12.2	49.5	11.6	**50.4**
	Ours	**18.0**	47.0	**14.5**	48.9	**16.5**	49.5	**15.2**	49.8
5	Faster R-CNN [9]	0	44.1	0	46.3	0	48.7	0	**51.8**
	RepMet [26]	11.3	49.4	5.6	**51.1**	5.7	49.6	11.6	49.2
	FeatReweighting [28]	12.1	**49.7**	7.0	46.8	13.7	47.9	6.8	48.5
	TFA [31]	16.2	48.8	3.6	49.5	13.8	**52.4**	9.6	49.6
	Meta R-CNN [29]	16.8	48.2	12.5	50.2	15.4	50.0	13.6	51.2
	Ours	**22.8**	48.4	**14.9**	49.1	**18.8**	49.9	**17.5**	49.9
10	Faster R-CNN [9]	0	44.2	1.8	43.2	0	49.1	0	50.8
	RepMet [26]	12.1	49.3	5.9	51.2	7.4	52.3	12.8	50.5
	FeatReweighting [28]	18.1	49.5	9.0	46.4	13.8	44.5	7.2	45.7
	TFA [31]	22.9	49.2	5.2	48.5	14.6	**52.8**	11.4	48.3
	Meta R-CNN [29]	22.0	49.4	15.3	50.3	20.3	52.2	17.0	**52.3**
	Ours	**27.6**	**50.9**	**18.9**	**52.5**	**23.3**	52.1	**18.9**	51.7
20	Faster R-CNN [9]	7.1	45.8	1.8	46.3	11.2	48.5	2.4	45.2
	RepMet [26]	12.2	48.6	6.8	50.4	7.8	52.4	16.7	**53.5**
	FeatReweighting [28]	22.0	50.0	14.1	43.5	18.5	43.5	12.2	44.4
	TFA [31]	23.7	49.1	7.8	48.5	18.9	52.4	12.8	50.2
	Meta R-CNN [29]	24.3	49.3	20.5	**52.3**	26.3	**53.7**	22.0	53.4
	Ours	**29.6**	**52.2**	**22.8**	51.6	**28.8**	53.1	**25.7**	52.3
30	Fatser R-CNN [9]	19.8	44.4	5.3	44.3	17.7	44.3	7.9	49.2
	RepMet [26]	15.1	50.9	6.5	50.7	8.6	50.7	17.3	51.2
	FeatReweighting [28]	29.6	50.4	14.4	44.4	23.1	45.2	15.9	41.7
	TFA [31]	24.0	50.6	9.8	49.2	19.9	52.6	14.5	50.5
	Meta R-CNN [29]	28.1	51.8	23.5	**53.2**	29.0	**54.6**	25.3	**53.7**
	Ours	**32.6**	**53.3**	**25.7**	53.0	**32.1**	53.6	**28.7**	53.6

The numbers of the images in the training set, validation set, and testing set are 5862, 5863, and 11 738, respectively. The spatial resolution ranges from 0.5 to 30 m and the size of images in this dataset is 800×800 pixels.

To establish the few-shot learning setup, we select five classes from its 20 object categories as the novel ones while keeping the remaining 15 classes as the base ones. We construct four different novel/base splits in which the novel categories do not overlap with each other. Table I presents the specific categories of each split. For training, we have $K = 3, 5, 10, 20$, and 30 object instances for each of the novel classes, which are randomly sampled from the combination of the training set and the validation set. We evaluate our method on the testing images, which contains both base and novel classes.

For evaluation metrics, we use the general object detection criterion, namely mean average precision (mAP). The PASCAL VOC2007 development kit is used to evaluate the performance. We run each experiment for three times to obtain relatively stable results.

B. Implementation Details

We construct our model and conduct our experiments based on the Pytorch [46] deep learning framework. A single GTX TITAN XP GPU is used for training and testing for acceleration. Since we implement our proposed P-CNN based on Faster R-CNN which employs ResNet101 [47] pretrained on ImageNet as the backbone network, our hyperparameters follow their original report. During the training process, we use an SGD optimizer with a base learning rate of 0.001, a momentum coefficient of 0.9, and a weight decay of 0.0001. For the first training phase, we train the model for 20 epochs. In the second phase, training the model for ten epochs is enough to obtain good performance on both base and novel classes. In the testing process, we directly load the prototypes of each class that are obtained by simply passing the support images to PLN without backpropagation. Thus, once we obtain the prototypes of all categories, the PLN can be removed. In each of our testing experiments, there are 20 prototypes containing 15 base classes and five novel classes for the DIOR dataset and we test all the 11 738 images in the testing set.

C. Experimental Results and Comparisons

To verify the effectiveness of our proposed P-CNN, we make comprehensive comparisons with four state-of-the-art few-shot object detection methods, including RepMet [26], FeatReweighting [28], a two-stage fine-tuning approach (TFA) [31], and Meta R-CNN [29], as well as the Faster R-CNN [9], which serves as the baseline of our method. The results are shown in Table II.

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From Table II, we have the following observations. First, for the Faster R-CNN [9], the results show that without sufficient number of bounding-box annotations, Faster R-CNN performs poorly to detect novel-class objects. It is worth noting that the performance on novel classes is extremely bad, which indirectly demonstrates that our two-phase training strategy can effectively alleviate the overfitting problem. Second, compared to RepMet [26], our method shows superior performance in most cases, especially for novel classes. In addition, when the number of shots increases, the performance improvements of the novel classes are significantly higher than those of the RepMet [26]. Third, with respect to FeatReweighting [28], we find that P-CNN surpasses it in all novel classes and most cases of base class. Note that, with the number of samples grows, the performance of FeatReweighting for novel classes gradually gets better. However, the performance of some base classes becomes worse. We attribute this phenomenon to the change of weight distribution for the classifier. As illustrated in [48], the classifier weight norm is related to the number of training instances. The classifiers for the base classes would see more and diverse training instances, leading to dominating weight magnitude, which would make the classification scores for base classes much bigger than those of novel classes. When more instances for novel classes are obtained, such weight distribution has been gradually changed, leading to the improvement of the novel classes and the degeneration of the base classes. Fourth, as for TFA [31], although the accuracy of some base classes is high (e.g., base classes in split 3), there also exist some novel classes with extremely low detection accuracy (e.g., novel classes in split 3), which results in poor overall detection results. In contrast, our results are relatively evenly distributed among both novel and base classes. Finally, in comparison with Meta R-CNN [29] that is a strong competitor, without bells and whistles, our model performs better than it in most cases. The main difference between our model and Meta R-CNN is that we introduce the P-G RPN module to provide better foreground proposals, which is especially important to facilitate subsequent classification and localization. We can observe that the improvement is very small for the base set because there are enough training examples to train the model, whereas significant improvement is obtained on the novel set due to the guidance of prototypes for RPN.

Qualitatively, we visualize the results of 30-shot model in the first base/novel split, as shown in Fig. 4. Our proposed P-CNN is able to simultaneously detect both novel and base objects. We show the success cases for the images containing novel classes, base classes, and both novel and base classes, as well as some failure cases. The first two rows present the detection results for novel and base classes respectively. The third row shows more challenging situation where both novel and base categories exist, which fully demonstrates the effectiveness of our model. Besides, in the last row, we show some typical failure cases, which can be roughly divided into three types: 1) between-class similarity; 2) dense distribution; and 3) ambient occlusion. More specifically, since the model is trained by many vehicle examples and the ships and vehicles seem similar in aspect ratio, size, and appearance in remote sensing images, the ships scattered on the rivers tend to be misclassified as vehicles. This problem also exists between overpasses and bridges. The dense distribution of basketball court makes it difficult to accurately detect the instances. In the second image of the last row, one chimney is missed because of the occlusion of smoke. These aforementioned problems demand further research.

TABLE III

ABLATION STUDIES FOR THE FOUR BASE/NOVEL SPLITS. WE USE MAP AS EVALUATION MEASURE AND THE BEST RESULTS ARE INDICATED IN BOLD FONT

#shots	Instance patch	R-G RPN	Soft-attention	Split 1	Split 2	Split 3	Split 4
3		√	√	14.2	10.9	12.1	11.8
	√		√	14.1	11.2	13.6	12.0
	√	√		12.8	10.2	11.4	10.7
	√	√	√	**16.7**	**12.0**	**14.9**	**13.9**
5		√	√	18.8	13.0	16.0	13.6
	√		√	16.1	11.9	15.4	14.1
	√	√		18.3	11.4	14.6	13.3
	√	√	√	**21.4**	**13.9**	**17.4**	**14.6**
10		√	√	23.3	14.9	21.0	18.5
	√		√	23.7	15.8	22.1	17.0
	√	√		21.8	**16.5**	18.2	15.1
	√	√	√	**26.1**	16.3	**22.8**	**18.6**
20		√	√	27.5	20.8	25.8	23.1
	√		√	27.9	20.3	24.7	22.5
	√	√		26.0	18.9	25.0	19.7
	√	√	√	**28.8**	**21.9**	**27.6**	**24.1**
30		√	√	28.9	23.9	29.6	25.7
	√		√	30.1	22.4	28.7	26.4
	√	√		28.5	24.0	27.9	25.8
	√	√	√	**32.3**	**25.7**	**31.7**	**27.6**

D. Ablation Studies

We perform ablation studies to demonstrate the rationality and effectiveness of our input format, the P-G RPN module, and the channelwise soft-attention operation for the modification of the detection head. These experiments are conducted on different shots and different base/novel split settings, and the results are reported in Tables III and IV.

1) Analysis of Different Input Formats: In some few-shot object detection works for natural scene images [28], [29], the input format of support set is four channels, which concatenates each three-channel RGB image with an additional one-channel binary mask for the object bounding box. The mask provides clues for the network to locate the position of the object instance, which is helpful for detection task. Then, the four-channel inputs are further resized to a smaller size to ease computation burden. However, such setting might be unsuitable for remote sensing images because it is often the case that the object instances in the remote sensing images only account for a small proportion of the entire images. Thus, the information of prototypes is easily distorted. To this end, we abandon using the mask as input and directly crop the object instances from the original remote sensing images instead. Then, the input image patches of support set are resized to 224×224. We observe that using instance patches as input brings considerable improvement.

2) Analysis of the P-G RPN Module: The goal of P-G RPN is to provide better proposals. We conduct comparison

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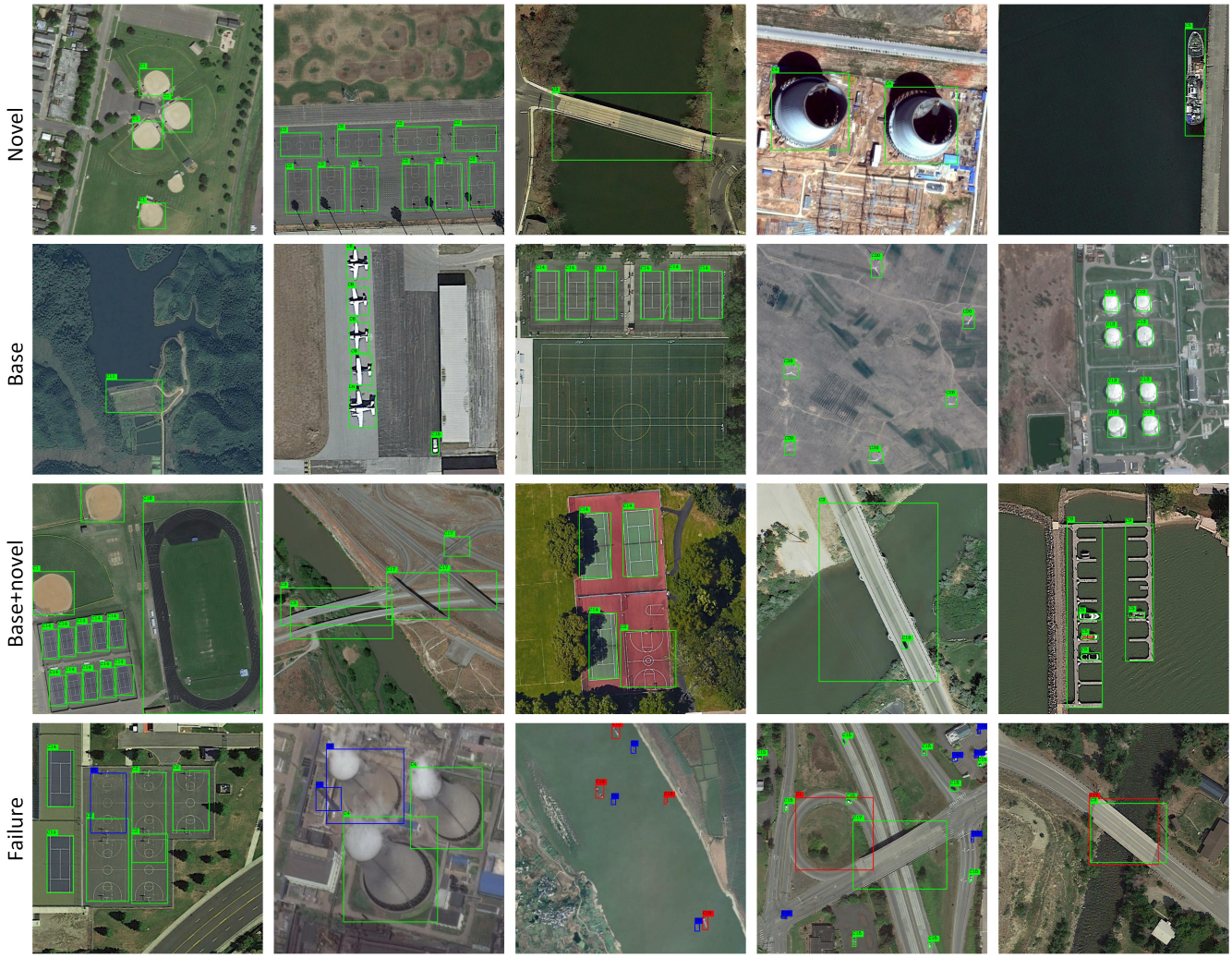


Fig. 4. Visualization of detection results. The first two rows present the results for novel and base classes, respectively, the third row shows more challenging situation where both novel and base categories exist, and the last row shows some typical failure cases. Green, blue, and red boxes denote true positive, false negative, and false positive, respectively.

TABLE IV

ABLATION STUDIES OF MULTIORIENTATION SAMPLE AUGMENTATION. "W/O M-O" IS FOR "WITHOUT MULTIORIENTATION" AND "W M-O" IS FOR "WITH MULTIORIENTATION". WE USE MAP AS EVALUATION MEASURE AND THE BEST RESULTS ARE INDICATED IN BOLD FONT

	Split 1		Split 2		Split 3		Split 4	
#shots	w/o M-O	w M-O	w/o M-O	w M-O	w/o M-O	w M-O	w/o M-O	w M-O
3 shots	16.7	**18.0**	12.0	**14.5**	14.9	**16.5**	13.9	**15.2**
5 shots	21.4	**22.8**	13.9	**14.9**	17.4	**18.8**	14.6	**17.5**
10 shots	26.1	**27.6**	16.3	**18.9**	22.8	**23.3**	18.6	**18.9**
20 shots	28.8	**29.6**	21.9	**22.8**	27.6	**28.8**	24.1	**25.7**
30 shots	32.3	**32.6**	25.7	**25.7**	31.7	**32.1**	27.6	**28.7**

experiments without using the P-G RPN module. Obviously, our proposed P-CNN with P-G RPN performs better.

3) Analysis of the Channelwise Soft-Attention Operation: The channelwise soft-attention operation is widely used in few-shot object detection [28], [29], which would greatly improve the detection performance by refining the query features to be detected. Our experiments show that this operation also works well on remote sensing images.

4) Analysis of the Multiorientation Sample Augmentation: The varied object orientations bring a big challenge to object detection in remote sensing images especially in the case of few-shot samples. In the few-shot scenario, the objects with different orientations are not accessible. To this end, multiorientation sample augmentation scheme can enrich the orientation space. Without bells and whistles, considerable gains are obtained.

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VI. Conclusion

In this article, for the purpose of adapting the general object detection framework of Faster R-CNN to few-shot object detection task in remote sensing images, we proposed a novel P-CNN method based on the core idea that prototypes can provide important clues for detection. PLN is applied on support image patches to learn the prototypes of each class. To fully mine the information of support set, the prototypes are used in the following P-G RPN and detection head modules. In the P-G RPN model, we use prototypes to guide RPN to generate higher quality candidate boxes, which is especially beneficial for few-shot object detection in remote sensing images to select foreground targets from the complex background. In the detection head module, prototypes exhibit their extraordinary guiding ability to remodel the features of RoIs, which can further boost the detection performance.

To alleviate the challenge of small training samples, leveraging data augmentation is a promising direction for few-shot learning. In the future, we will carry out follow-up research by using some interesting data augmentation techniques such as cut-and-paste [49] and copy-and-paste [50].

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